

HALF - WAVE RECTIFIER



AIM:

To rectify the AC signal and then to find out the Ripple factor and percentage of Regulation in Half-Wave Rectifier with and without Capacitor filter using : LTSpice Simulator.



SOFTWARE REQUIRED:

- ▶ LTSpice Simulator.



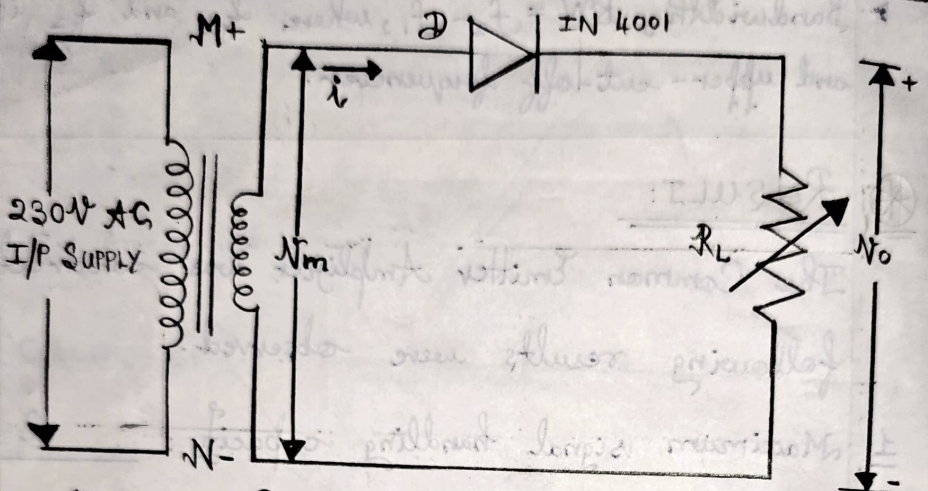
PROCEDURE:

1. Connecting the circuit in LTSpice workspace as per the given circuit diagram.
2. Connect the primary of the transformer to main supply, i.e, 230V, 50Hz.
3. Connect the decade resistance box and set the R_L value to 100Ω.
4. Disconnect load resistance (DRB) and note down no load voltage V_{dc} ($V_{no\ load}$).
5. Connect load resistance at 1kΩ and connect Channel-II of CRO at output terminals and CH-I of CRO at Secondary Input terminals.
6. Observe and note down the Input and Output Wave Form on the graph sheet.

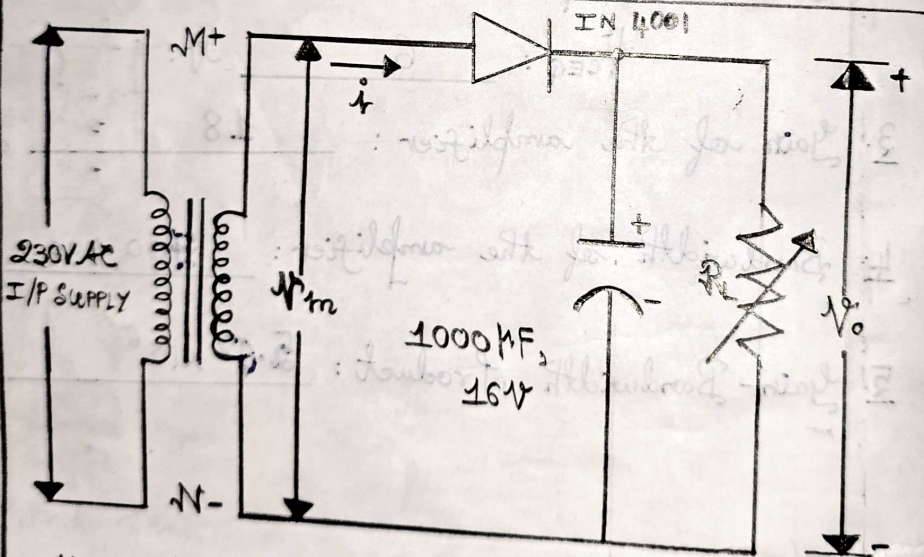


CIRCUIT DIAGRAM:

WITHOUT CAPACITOR FILTER AND WITH CAPACITOR FILTER:



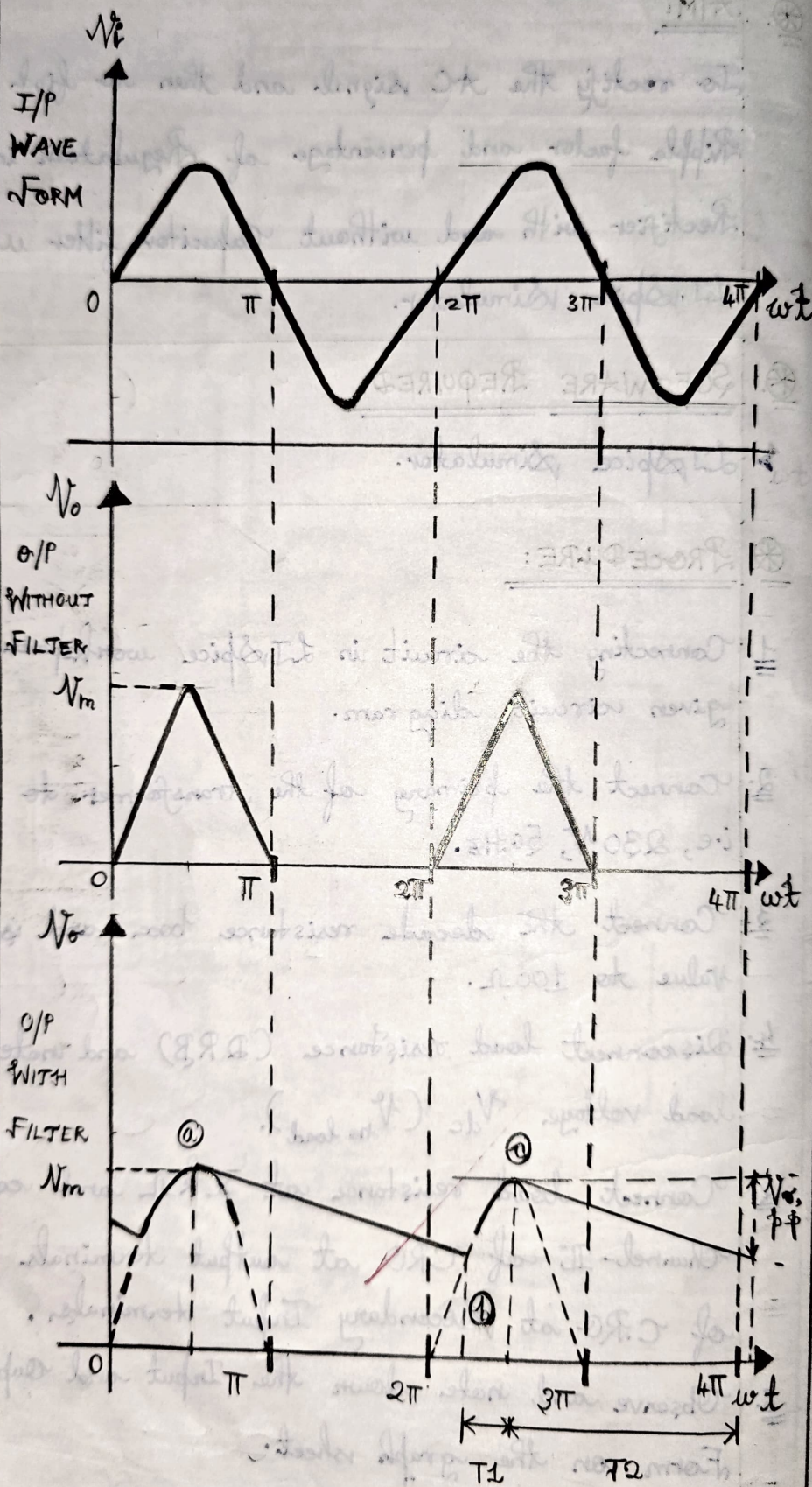
HALF-WAVE RECTIFIER WITHOUT FILTER



HALF-WAVE RECTIFIER WITH CAPACITOR FILTER



WAVE SHAPES:



HALF-WAVE RECTIFIER WITH CAPACITOR FILTER WAVE FORM



TABULATION:

WITHOUT FILTER:

(V) No LOAD VOLTAGE : $(V_{NL}) = 15.8 \text{ V}$

S.No.	LOAD RESISTANCE $R_L (\Omega)$	O/P VOLTAGE (V_o)		RIPPLE FACTOR $(\gamma = \frac{V_{ac}}{V_{dc}})$	% OF REGULATION $\left(\frac{V_{NL} - V_{FL}}{V_{NL}} \right) \times 100$
		$V_{ac} (V)$	$V_{dc} (V)$		
1.	100	6.2	6.5	0.95	58.86
2.	200	5.8	7.8	0.74	50.63
3.	300	5.5	8.9	0.62	43.67
4.	400	5.2	9.8	0.53	38.00
5.	500	5.0	10.6	0.47	32.91
6.	600	4.8	11.3	0.42	28.48
7.	700	4.6	11.9	0.39	24.68
8.	800	4.5	12.4	0.36	21.52
9.	900	4.3	12.8	0.34	18.99
10.	1000	4.2	13.2	0.32	16.46

WITH CAPACITOR FILTER:

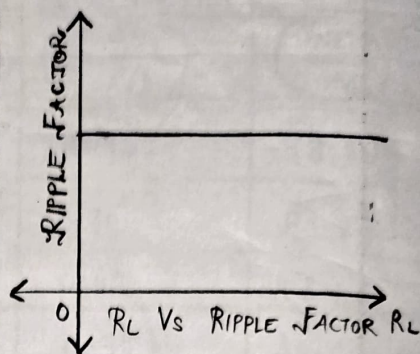
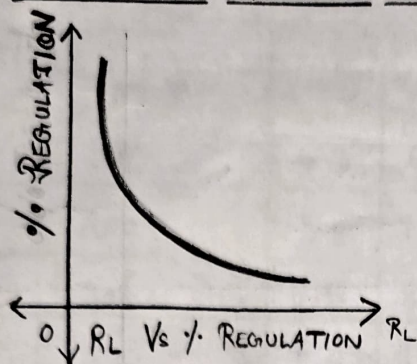
(V) No LOAD VOLTAGE : $(V_{NL}) = 16.5 \text{ V}$

S.No.	LOAD RESISTANCE $R_L (\Omega)$	O/P VOLTAGE (V_o)		RIPPLE FACTOR $(\gamma = \frac{V_{ac}}{V_{dc}})$	% OF REGULATION $\left(\frac{V_{NL} - V_{FL}}{V_{NL}} \right) \times 100$
		$V_{ac} (V)$	$V_{dc} (V)$		
1.	100	1.5	12.5	0.12	24.24
2.	200	1.3	13.2	0.10	20.00
3.	300	1.2	13.8	0.087	16.36
4.	400	1.1	14.3	0.077	13.33
5.	500	1.0	14.7	0.068	10.91
6.	600	0.9	15.0	0.06	9.09
7.	700	0.85	15.3	0.055	7.27
8.	800	0.8	15.5	0.051	6.06
9.	900	0.75	15.7	0.047	4.85
10.	1000	0.7	15.9	0.044	3.64

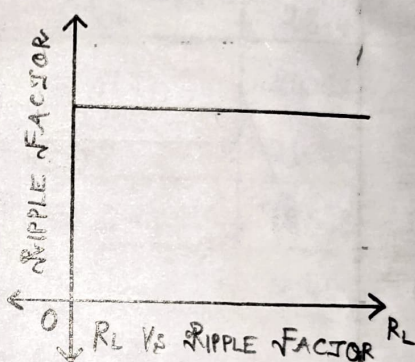
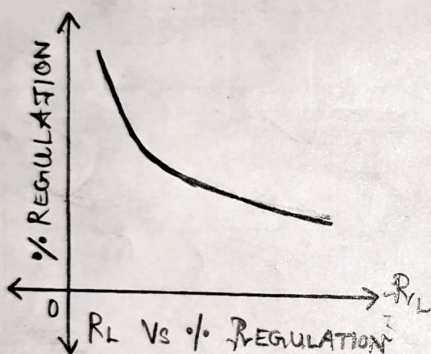


MODEL GRAPHS:

WITHOUT FILTER WAVE FORM:



WITH FILTER WAVE FORM:



CALCULATION:

i). RIPPLE FACTOR:

$$\gamma = \frac{V_{ac}}{V_{dc}}$$

$$\gamma = \frac{6.2}{6.5} = 0.95$$

[Example: Without filter;
 $R_L = 100 \Omega$]

ii). PERCENTAGE REGULATION:

$$\% \text{ (Regulation)} = \frac{V_{NL} - V_{FL}}{V_{NL}} \times 100\%$$

$$\Rightarrow \xi: [\text{Without filter, } R_L = 100 \Omega] = \frac{15.8 - 6.5}{15.8} \times 100\% = 58.86\%$$

RESULT:

Observe Input and Output Wave Forms and calculate the Ripple Factor and the percentage (%) of Regulation in Half-Wave Rectifier with and without filter.

► Without filter:-

- Ripple Factor: 0.32 to 0.95 .
- Regulation: 16.46% to 58.86% .

► With Capacitor Filter:

- Ripple Factor: 0.044 to 0.12 .
- Regulation: 3.64% to 24.24% .

FULL-WAVE RECTIFIERS



AIM:

To rectify the AC signal and then to find out the Ripple Factor and the percentage of Regulation in Full-Wave Rectifier - center tapped circuit - with and without Capacitor filter.



SOFTWARE REQUIRED:

- ▶ LTSpice Simulator.



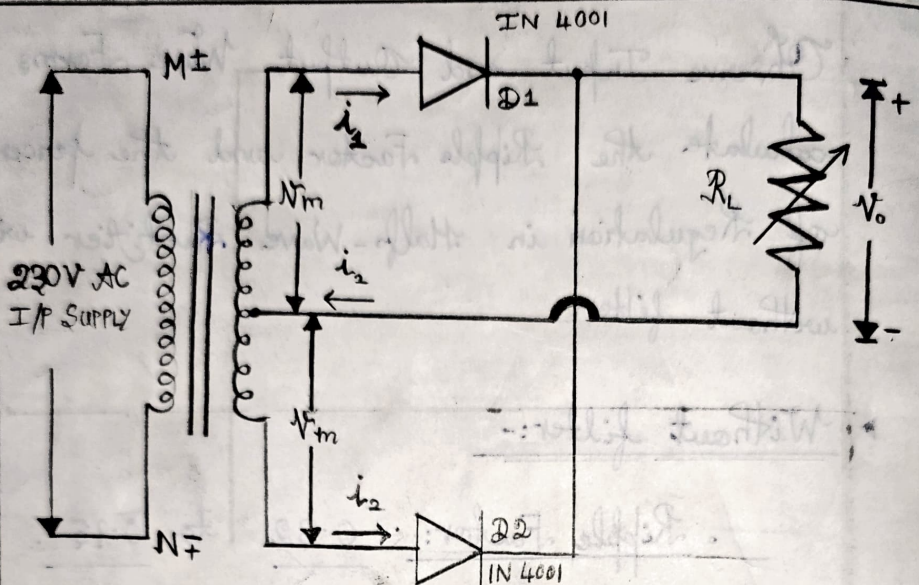
PROCEDURE:

1. Connecting the circuit on bread board as per the circuit diagram.
2. Connect the primary of the transformer to main supply, i.e., 230V, 50Hz.
3. Connect the decade resistance box and set the R_L value to 100Ω .
4. Connect the Multimeter at output terminals and vary the load resistance (DRB).
5. Disconnect load resistance (DRB) and note down no load voltage V_{dc} ($V_{no\ load}$).
6. Connect load resistance at $1K\Omega$ and connect Channel-II of CRO at output terminals and CH-I of CRO at Secondary Input terminals.
7. Observe and note down the Input and Output Wave

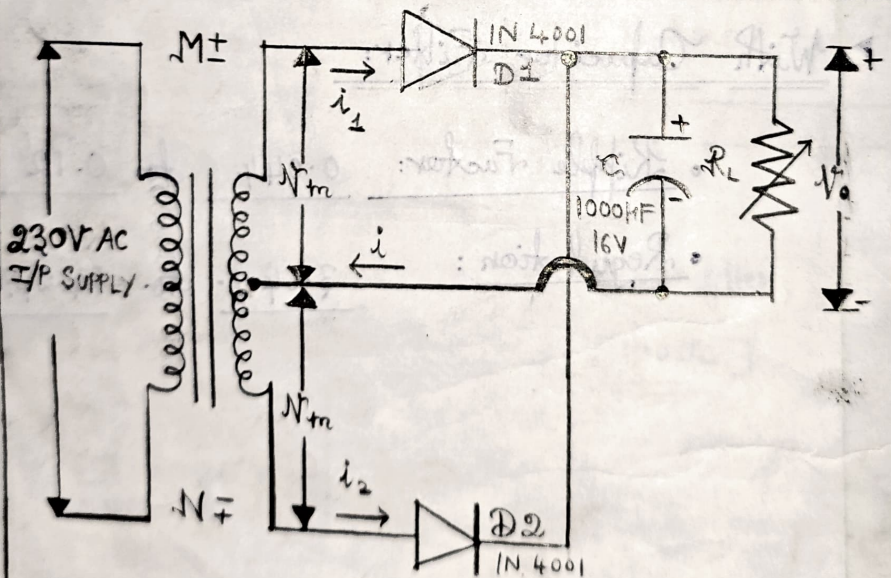


CIRCUIT DIAGRAM:

WITHOUT FILTER AND WITH FILTER:



Full-Wave Rectifier Without Filter



Full-Wave Rectifier With Capacitor Filter



Form on the graph sheet.

8. Calculate the Ripple Factor: $\gamma = \frac{V_{ac}}{V}$

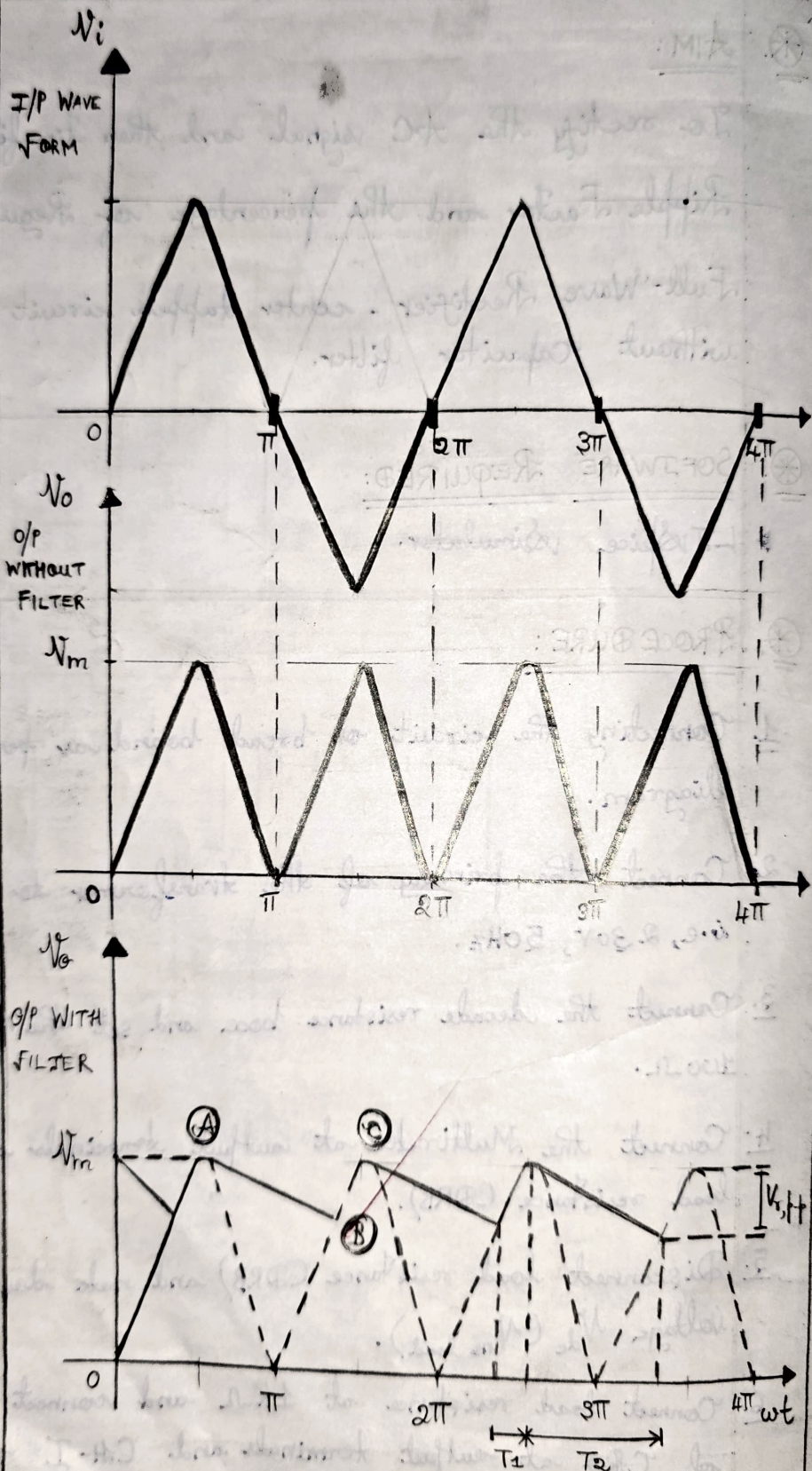
100	0.01	1.0	0.01
200	0.02	1.0	0.02
300	0.03	1.0	0.03
400	0.04	1.0	0.04
500	0.05	1.0	0.05
600	0.06	1.0	0.06
700	0.07	1.0	0.07
800	0.08	1.0	0.08
900	0.09	1.0	0.09
1000	0.10	1.0	0.10

With Capacitor Filter:

Capacitor Value (μF)	Output Voltage (V)	Ripple Factor (%)	Regulation (%)
100	1.0	1.0	1.0
200	1.1	1.1	1.1
300	1.2	1.2	1.2
400	1.3	1.3	1.3
500	1.4	1.4	1.4
600	1.5	1.5	1.5
700	1.6	1.6	1.6
800	1.7	1.7	1.7
900	1.8	1.8	1.8
1000	1.9	1.9	1.9



WAVE SHAPES:



FULL-WAVE RECTIFIER WITH CAPACITOR FILTER
WAVE FORM



TABULATION:

WITHOUT FILTER:

(V) NO LOAD VOLTAGE: $(V_{dc}) = 14.2 \text{ V}$.

S.No.	LOAD RESISTANCE $R_L (\Omega)$	O/P VOLTAGE (V_o)		RIPPLE FACTOR ($\gamma = \frac{V_{ac}}{V_{dc}}$)	% OF REGULATION ($\frac{V_{NL} - V_{FL} \times 100}{V_{NL}}$)
		$V_{ac}(V)$	$V_{dc}(V)$		
1.	100	4.8	7.5	0.64	47.18
2.	200	4.5	8.8	0.51	38.03
3.	300	4.2	9.9	0.42	30.28
4.	400	4.0	10.8	0.37	23.94
5.	500	3.8	11.6	0.33	18.31
6.	600	3.6	12.2	0.30	14.08
7.	700	3.4	12.7	0.27	10.56
8.	800	3.3	13.1	0.25	7.75
9.	900	3.2	13.4	0.24	5.63
10.	1000	3.0	13.7	0.22	3.52

WITH CAPACITOR FILTER:

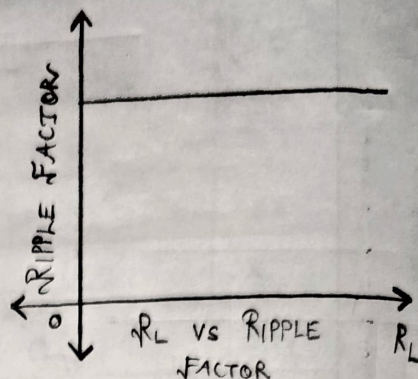
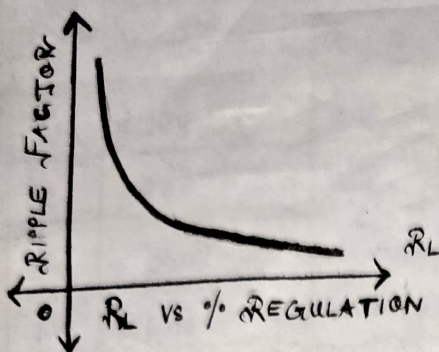
(V) NO LOAD VOLTAGE: $(V_{dc}) = 16.8 \text{ V}$.

S.No.	LOAD RESISTANCE $R_L (\Omega)$	O/P VOLTAGE (V_o)		RIPPLE FACTOR ($\gamma = \frac{V_{ac}}{V_{dc}}$)	% OF REGULATION ($\frac{V_{NL} - V_{FL} \times 100}{V_{NL}}$)
		$V_{ac}(V)$	$V_{dc}(V)$		
1.	100	1.2	13.5	0.089	19.64
2.	200	1.1	14.2	0.077	15.48
3.	300	1.0	14.8	0.067	11.90
4.	400	0.95	15.3	0.062	8.93
5.	500	0.9	15.7	0.057	6.55
6.	600	0.85	16.0	0.053	4.76
7.	700	0.8	16.2	0.049	3.57
8.	800	0.75	16.4	0.046	2.38
9.	900	0.7	16.6	0.042	1.19
10.	1000	0.65	16.7	0.039	0.60

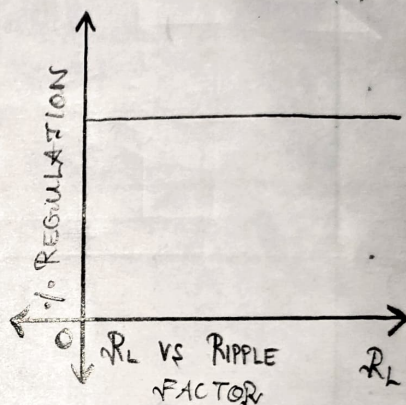
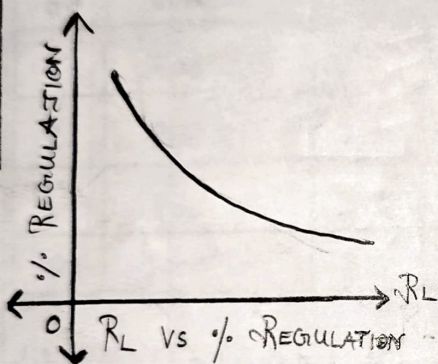


MODEL GRAPHS:

WITHOUT FILTER WAVE FORM:



WITH FILTER WAVE FORM:



CALCULATION:

i) Ripple Factor:

$$\gamma = \frac{V_{ac}}{V_{dc}}$$

$$\Rightarrow \gamma = \frac{4.8}{7.5} = 0.64$$

$\Rightarrow$ Eg: (Without filter; $R_L = 100 \Omega$):

ii) Percentage Regulation:

$$\% \text{ (Regulation)} = \frac{V_{NL} - V_{FL}}{V_{NL}} \times 100\%$$

$$\begin{aligned} \Rightarrow & \frac{14.2 - 7.5}{14.2} \times 100\% \\ & = 47.18\% \end{aligned}$$

$\rightarrow$ Eg: (Without filter; $R_L = 100 \Omega$):

RESULT:

Observe Input and Output Wave forms and calculate the Ripple factor and percentage (%) of Regulation in Full-Wave Rectifier with and without filter.

Without Filter:

- Ripple Factor: $\underline{0.22}$ to $\underline{0.64}$ %
- Regulation: $\underline{3.52\%}$ to $\underline{47.18\%}$ %

With Capacitor Filter:

- Ripple Factor: $\underline{0.039}$ to $\underline{0.089}$ %
- Regulation: $\underline{0.60\%}$ to $\underline{19.64\%}$ %